```
<!DOCTYPE html>
<html lang="en-UK">
<head>
<meta charset="UTF-8" />
<meta name="generator" content="LaTeX Lwarp package" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Sample_Lwarp_Xy-jax</title>
<link rel="stylesheet" type="text/css" href="my_file_stand-alone.css" />

<!-- Joao Faria Martins, 1st Nov 2022:
Original file called
lwarp_mathjax.txt
provided by lwarpmk: v0.902.
File tweaked by Joao Faria Martins, 7 January 2022 and 8 January 2022.
Additions to Lwarp_mathjax.txt:
Two lines were added to the script, as shown,  in order that browsers can find Xy-Jaxv3, hopefully without breaking the MathJax emulation code.
In particular these additions to the original configuration do not seem to break the equation numbering.

Lots of credit to instructions in https://github.com/sonoisa/XyJax-v3 and
in "40.11 lwarp_mathjax.txt" in the official Lwarp manual, http://mirrors.ibiblio.org/CTAN/macros/latex/contrib/lwarp/lwarp.pdf
--

This modification in the Lwarp configuration file was produced under items 5 and 6 of the licence https://www.ctan.org/license/lppl1.3
and also the following paragraph in the official Lwarp manual, http://mirrors.ibiblio.org/CTAN/macros/latex/contrib/lwarp/lwarp.pdf
"40.11 lwarp_mathjax.txt custom script
To generate a custom script, such as to use a local repository, copy lwarp_mathjax.txt to a new file, make changes while keeping lwarp-s adjustments for equation numbering and starred macros, and use \MathJaxFilename to select the new filename."
-->



<script>
// Lwarp MathJax emulation code
//
// Based on code by Davide P. Cervone.
// Equation numbering: https://github.com/mathjax/MathJax/issues/2427
// Starred and ifnextchar macros: https://github.com/mathjax/MathJax/issues/2428
// \left, \right delimiters: https://github.com/mathjax/MathJax/issues/2535
//
// Modified by Brian Dunn to adjust equation numbering and add subequations.
//
// LaTeX can use \seteqnumber{subequations?}{section}{number} before each equation.
// subequations? is 0 usually, 1 if inside subequations.
// section is a string printed as-is, or empty.
// number is auto-incremented by MathJax between equations.
//
// Tweaks by Joao Faria Martins (08/01/2022) in order that Xy-jaxv3 can be found. Lots of credit to instructions in https://github.com/sonoisa/XyJax-v3
// added by Joao Faria Martins: 07 Jan 2022  '[custom]/xypic.js',  paths: {custom: 'https://cdn.jsdelivr.net/gh/sonoisa/XyJax-v3@3.0.1/build/'}
// Corrected by Joao Faria Martins: 25 May 2026  '[custom]/xypic.js',  paths: {custom: 'https://cdn.jsdelivr.net/gh/sonoisa/XyJax-v3@3.0.1/build/'} should be:
//paths: {      custom: './js/'   } and  '[custom]/xypic'. This works for a local copy of xyjax.


MathJax = {
     subequations: "0",
     section: "",
     loader: {
          load: ['[tex]/tagformat', '[tex]/textmacros', '[custom]/xypic'],

      paths: {
      custom: './js/'
             }


     },
     startup: {
          ready() {
              //       These would be replaced by import commands if you wanted to make
              //       a proper extension.
              const Configuration = MathJax._.input.tex.Configuration.Configuration;
              const CommandMap = MathJax._.input.tex.SymbolMap.CommandMap;
              const Macro = MathJax._.input.tex.Symbol.Macro;
              const TexError = MathJax._.input.tex.TexError.default;
              const ParseUtil = MathJax._.input.tex.ParseUtil.default;
              const expandable = MathJax._.util.Options.expandable;


              //       Insert the replacement string into the TeX string, and check
              //       that there haven't been too many maxro substitutions (prevents
              //       infinite loops).
              const useArgument = (parser, text) => {
                   parser.string = ParseUtil.addArgs(parser, text, parser.string.slice(parser.i));
                   parser.i = 0;
                   if (++parser.macroCount > parser.configuration.options.maxMacros) {
                        throw new TexError('MaxMacroSub1',
                        'MathJax maximum macro substitution count exceeded; ' +
                        'is there a recursive macro call?');
                   }
              }


              //       Create the command map for:
              //            \ifstar, \ifnextchar, \ifblank, \ifstrequal, \seteqnumber
              new CommandMap('Lwarp-macros', {
                   ifstar: 'IfstarFunction',
                   ifnextchar: 'IfnextcharFunction',
                   ifblank: 'IfblankFunction',
                   ifstrequal: 'IfstrequalFunction',
                   seteqnumber: 'SeteqnumberFunction'
              }, {
                   //       This function implements an ifstar macro.
                   IfstarFunction(parser, name) {
                        const resultstar = parser.GetArgument(name);
                        const resultnostar = parser.GetArgument(name);
                        const star = parser.GetStar();                         // true if there is a *
                        useArgument(parser, star ? resultstar : resultnostar);
                   },


                   //       This function implements an ifnextchar macro.
                   IfnextcharFunction(parser, name) {
                        let whichchar = parser.GetArgument(name);
                        if (whichchar.match(/^(?:0x[0-9A-F]+|[0-9]+)$/i)) {
                            // $ syntax highlighting
                            whichchar = String.fromCodePoint(parseInt(whichchar));
                        }
                        const resultnextchar = parser.GetArgument(name);
                        const resultnotnextchar = parser.GetArgument(name);
                        const gotchar = (parser.GetNext() === whichchar);
                        useArgument(parser, gotchar ? resultnextchar : resultnotnextchar);
                   },


                   // This function implements an ifblank macro.
                   IfblankFunction(parser, name) {
                        const blankarg = parser.GetArgument(name);
                        const resultblank = parser.GetArgument(name);
                        const resultnotblank = parser.GetArgument(name);
                        const isblank = (blankarg.trim() == "");
                        useArgument(parser, isblank ? resultblank : resultnotblank);
                   },


                   // This function implements an ifstrequal macro.
                   IfstrequalFunction(parser, name) {
                        const strequalfirst = parser.GetArgument(name);
                        const strequalsecond = parser.GetArgument(name);
                        const resultequal = parser.GetArgument(name);
                        const resultnotequal = parser.GetArgument(name);
                        const isequal = (strequalfirst == strequalsecond);
                        useArgument(parser, isequal ? resultequal : resultnotequal);
                   },


                   //       This function modifies the equation numbers.
                   SeteqnumberFunction(parser, name) {
                            //   Get the macro parameters
                            const star = parser.GetStar();                       // true if there is a *
                            const optBrackets = parser.GetBrackets(name);        // contents of optional brackets
                            const newsubequations = parser.GetArgument(name);      // the subequations argument
                            const neweqsection = parser.GetArgument(name);      // the eq section argument
                            const neweqnumber = parser.GetArgument(name);        // the eq number argument
                            MathJax.config.subequations=newsubequations ;       // a string with boolean meaning
                            MathJax.config.section=neweqsection ;               // a string with numeric meaning
                            parser.tags.counter = parser.tags.allCounter = neweqnumber ;
                   }


              });


              //       Create the Lwarp-macros package
              Configuration.create('Lwarp-macros', {
                   handler: {macro: ['Lwarp-macros']}
              });


              MathJax.startup.defaultReady();


              // For forward references:
              MathJax.startup.input[0].preFilters.add(({math}) => {
                   if (math.inputData.recompile){
                            MathJax.config.subequations = math.inputData.recompile.subequations;
                            MathJax.config.section = math.inputData.recompile.section;
                   }
              });
              MathJax.startup.input[0].postFilters.add(({math}) => {
                   if (math.inputData.recompile){
                            math.inputData.recompile.subequations = MathJax.config.subequations;
                            math.inputData.recompile.section = MathJax.config.section;
                   }
              });


                   // For \left, \right with unicode-math:
                   const {DelimiterMap} = MathJax._.input.tex.SymbolMap;
                   const {Symbol} = MathJax._.input.tex.Symbol;
                   const {MapHandler} = MathJax._.input.tex.MapHandler;
                   const delimiter = MapHandler.getMap('delimiter');
                   delimiter.add('\\lBrack', new Symbol('\\lBrack', '\u27E6'));
                   delimiter.add('\\rBrack', new Symbol('\\rBrack', '\u27E7'));
                   delimiter.add('\\lAngle', new Symbol('\\lAngle', '\u27EA'));
                   delimiter.add('\\rAngle', new Symbol('\\rAngle', '\u27EB'));
                   delimiter.add('\\lbrbrak', new Symbol('\\lbrbrak', '\u2772'));
                   delimiter.add('\\rbrbrak', new Symbol('\\rbrbrak', '\u2773'));
                   delimiter.add('\\lbag', new Symbol('\\lbag', '\u27C5'));
                   delimiter.add('\\rbag', new Symbol('\\rbag', '\u27C6'));
                   delimiter.add('\\llparenthesis', new Symbol('\\llparenthesis', '\u2987'));
                   delimiter.add('\\rrparenthesis', new Symbol('\\rrparenthesis', '\u2988'));
                   delimiter.add('\\llangle', new Symbol('\\llangle', '\u2989'));
                   delimiter.add('\\rrangle', new Symbol('\\rrangle', '\u298A'));
                   delimiter.add('\\Lbrbrak', new Symbol('\\Lbrbrak', '\u27EC'));
                   delimiter.add('\\Rbrbrak', new Symbol('\\Rbrbrak', '\u27ED'));
                   delimiter.add('\\lBrace', new Symbol('\\lBrace', '\u2983'));
                   delimiter.add('\\rBrace', new Symbol('\\rBrace', '\u2984'));
                   delimiter.add('\\lParen', new Symbol('\\lParen', '\u2985'));
                   delimiter.add('\\rParen', new Symbol('\\rParen', '\u2986'));
                   delimiter.add('\\lbrackubar', new Symbol('\\lbrackubar', '\u298B'));
                   delimiter.add('\\rbrackubar', new Symbol('\\rbrackubar', '\u298C'));
                   delimiter.add('\\lbrackultick', new Symbol('\\lbrackultick', '\u298D'));
                   delimiter.add('\\rbracklrtick', new Symbol('\\rbracklrtick', '\u298E'));
                   delimiter.add('\\lbracklltick', new Symbol('\\lbracklltick', '\u298F'));
                   delimiter.add('\\rbrackurtick', new Symbol('\\rbrackurtick', '\u2990'));
                   delimiter.add('\\langledot', new Symbol('\\langledot', '\u2991'));
                   delimiter.add('\\rangledot', new Symbol('\\rangledot', '\u2992'));
                   delimiter.add('\\lparenless', new Symbol('\\lparenless', '\u2993'));
                   delimiter.add('\\rparengtr', new Symbol('\\rparengtr', '\u2994'));
                   delimiter.add('\\Lparengtr', new Symbol('\\Lparengtr', '\u2995'));
                   delimiter.add('\\Rparenless', new Symbol('\\Rparenless', '\u2996'));
                   delimiter.add('\\lblkbrbrak', new Symbol('\\lblkbrbrak', '\u2997'));
                   delimiter.add('\\rblkbrbrak', new Symbol('\\rblkbrbrak', '\u2998'));
                   delimiter.add('\\lvzigzag', new Symbol('\\lvzigzag', '\u29D8'));
                   delimiter.add('\\rvzigzag', new Symbol('\\rvzigzag', '\u29D9'));
                   delimiter.add('\\Lvzigzag', new Symbol('\\Lvzigzag', '\u29DA'));
                   delimiter.add('\\Rvzigzag', new Symbol('\\Rvzigzag', '\u29DB'));
                   delimiter.add('\\lcurvyangle', new Symbol('\\lcurvyangle', '\u29FC'));
                   delimiter.add('\\rcurvyangle', new Symbol('\\rcurvyangle', '\u29FD'));
                   delimiter.add('\\Vvert', new Symbol('\\Vvert', '\u2980'));
          }       // ready
     },           // startup

//xypic added: Joao Faria Martins (8 Jan 2022)
     tex: {
          packages: {'[+]': ['tagformat', 'Lwarp-macros', 'textmacros', 'xypic']},
          tags: "ams",
                   tagformat: {
                            number: function (n) {
                                 if(MathJax.config.subequations==0)
                                        return(MathJax.config.section + n);
                                 else
                                        return(MathJax.config.section + String.fromCharCode(96+n));
                            },
                   },
     }
}
</script>


<script
         id="MathJax-script"
         // src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-chtml.js" JFM 25May 2026 (Credit: copilot)
          src="https://cdn.jsdelivr.net/npm/mathjax@3.2.2/es5/tex-chtml.js"
></script>


</head>
<body>
<!--|Using lwarp|Sample_Lwarp_Xy-jax.html|-->


<div class="homepagegrid">


<main class="bodycontainer">


<section class="textbody">

<a id="Sample_Lwarp_Xy-jax-autofile-0"></a>

<!--MathJax customizations:-->
<div data-nosnippet
    style="display:none"
>

\(\newcommand{\footnotename}{footnote}\)

\(\def \LWRfootnote {1}\)

\(\newcommand {\footnote }[2][\LWRfootnote ]{{}^{\mathrm {#1}}}\)

\(\newcommand {\footnotemark }[1][\LWRfootnote ]{{}^{\mathrm {#1}}}\)

\(\let \LWRorighspace \hspace \)

\(\renewcommand {\hspace }{\ifstar \LWRorighspace \LWRorighspace }\)

\(\newcommand {\TextOrMath }[2]{#2}\)

\(\newcommand {\mathnormal }[1]{{#1}}\)

\(\newcommand \ensuremath [1]{#1}\)

\(\newcommand {\LWRframebox }[2][]{\fbox {#2}} \newcommand {\framebox }[1][]{\LWRframebox } \)

\(\newcommand {\setlength }[2]{}\)

\(\newcommand {\addtolength }[2]{}\)

\(\newcommand {\setcounter }[2]{}\)

\(\newcommand {\addtocounter }[2]{}\)

\(\newcommand {\arabic }[1]{}\)

\(\newcommand {\number }[1]{}\)

\(\newcommand {\noalign }[1]{\text {#1}\notag \\}\)

\(\newcommand {\cline }[1]{}\)

\(\newcommand {\directlua }[1]{\text {(directlua)}}\)

\(\newcommand {\luatexdirectlua }[1]{\text {(directlua)}}\)

\(\newcommand {\protect }{}\)

\(\def \LWRabsorbnumber #1 {}\)

\(\def \LWRabsorbquotenumber &quot;#1 {}\)

\(\newcommand {\LWRabsorboption }[1][]{}\)

\(\newcommand {\LWRabsorbtwooptions }[1][]{\LWRabsorboption }\)

\(\def \mathchar {\ifnextchar &quot;\LWRabsorbquotenumber \LWRabsorbnumber }\)

\(\def \mathcode #1={\mathchar }\)

\(\let \delcode \mathcode \)

\(\let \delimiter \mathchar \)

\(\def \oe {\unicode {x0153}}\)

\(\def \OE {\unicode {x0152}}\)

\(\def \ae {\unicode {x00E6}}\)

\(\def \AE {\unicode {x00C6}}\)

\(\def \aa {\unicode {x00E5}}\)

\(\def \AA {\unicode {x00C5}}\)

\(\def \o {\unicode {x00F8}}\)

\(\def \O {\unicode {x00D8}}\)

\(\def \l {\unicode {x0142}}\)

\(\def \L {\unicode {x0141}}\)

\(\def \ss {\unicode {x00DF}}\)

\(\def \SS {\unicode {x1E9E}}\)

\(\def \dag {\unicode {x2020}}\)

\(\def \ddag {\unicode {x2021}}\)

\(\def \P {\unicode {x00B6}}\)

\(\def \copyright {\unicode {x00A9}}\)

\(\def \pounds {\unicode {x00A3}}\)

\(\let \LWRref \ref \)

\(\renewcommand {\ref }{\ifstar \LWRref \LWRref }\)

\( \newcommand {\multicolumn }[3]{#3}\)

\(\require {textcomp}\)

\(\newcommand {\intertext }[1]{\text {#1}\notag \\}\)

\(\let \Hat \hat \)

\(\let \Check \check \)

\(\let \Tilde \tilde \)

\(\let \Acute \acute \)

\(\let \Grave \grave \)

\(\let \Dot \dot \)

\(\let \Ddot \ddot \)

\(\let \Breve \breve \)

\(\let \Bar \bar \)

\(\let \Vec \vec \)

\(\newcommand {\C }{ \mathbb {C} }\)

\(\newcommand {\Cc }{ \mathcal {C} }\)

\(\newcommand {\Dc }{ \mathcal {D} }\)

\( \newcommand {\Grp }{ \mathrm {Grp} } \)

\(\newcommand {\Z }{ \mathbb {Z} }\)

\(\newcommand {\ra }[1]{\xrightarrow {#1}}\)

\(\DeclareMathOperator {\Sym }{Sym}\)

\(\def \id {\mathrm {id}}\)

\(\newcommand {\Q }{\mathbb {Q}}\)

\(\renewcommand {\a }{{\alpha }}\)

\(\renewcommand {\b }{{\beta }}\)

\(\def \g {\gamma }\)

\(\def \w {\omega }\)

\(\def \e {\epsilon }\)

\(\def \z {\zeta }\)

\(\def \d {\partial }\)

\(\newcommand {\tHpb }[3]{{\mathbf {\overline {2H}}^{#1}_{(#2,#3)}}}\)

\(\newcommand {\Hpb }{\mathbf {\overline {H}}}\)

\(\def \red {\textcolor {red}}\)

\(\def \green {\textcolor {green}}\)

\(\def \blue {\textcolor {blue}}\)
</div>
<a id="Sample_Lwarp_Xy-jax-autopage-1"></a>
```

<div class="titlepage">

<h1>Sample HTML file: produced with lwarp, with mathematical formulae displayed with MathJax, and xymatrix commutative diagrams displayed with XyJax-v3</h1>

<div class="author">

<div class="oneauthor">

<p>
Joa&#x0303;o Faria Martins
</p>
</div>

</div>

<div class="titledate">

<p>
Last modified: May 25, 2024
</p>
</div>

</div>

```
<!--
......  section Contents ......
-->
<h4 id="autosec-3">Contents</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-3"></a>



<nav class="toc">

<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-4" class="tocsection" >
<span class="sectionnumber">1</span>&#x2003;Context and instructions: using Lwarp to compile latex to HTML, so that xymatrix diagrams are displayed with Xy-Jax</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-11" class="tocsection" >
<span class="sectionnumber">2</span>&#x2003;Examples: xymatrix diagrams displayed with Xy-Jaxv3 using Lwarp</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-13" class="tocsection" >
<span class="sectionnumber">3</span>&#x2003;Issues when using XyJax-v3 and Lwarp for typesetting xymatrix diagrams</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-24" class="tocsection" >
<span class="sectionnumber">4</span>&#x2003;More general notes</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-41" class="tocsection" >
<span class="sectionnumber">5</span>&#x2003;Some examples of tikz and xymatrix diagrams compiled as svg images, with alternative text</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-42" class="tocsubsection" >
<span class="sectionnumber">5.1</span>&#x2003;An xymatrix diagram compiled as an image with alt text</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-45" class="tocsubsection" >
<span class="sectionnumber">5.2</span>&#x2003;A tikz image compiled as an image with alt text:</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-49" class="tocsubsection" >
<span class="sectionnumber">5.3</span>&#x2003;A tikz image compiled to an image with alt text, outside a float environment</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-53" class="tocsubsection" >
<span class="sectionnumber">5.4</span>&#x2003;A tikz image compiled to an image with alt text, outside a float environment, and with a caption</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-57" class="tocsubsection" >
<span class="sectionnumber">5.5</span>&#x2003;A tikz-cd diagram compiled into a figure with alt text: </a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-60" class="tocsection" >
<span class="sectionnumber">6</span>&#x2003;Sample mathematics</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-61" class="tocsubsection" >
<span class="sectionnumber">6.1</span>&#x2003;Adjoint functors and coadjoint functors via universal arrows</a>
</p>


<p>
<a href="Sample_Lwarp_Xy-jax.html#autosec-66" class="tocsubsection" >
<span class="sectionnumber">6.2</span>&#x2003;The Galois correspondence for \(f(t)=t^3-2\), over the rational field</a>
</p>
</nav>

<p>
\[\xymatrix @R=30pt@C=60pt {&amp;A\ar [d]_{\iota _0^A} \ar [r]^f &amp; X\ar [d]_{\iota _0^X} \ar [r]^g &amp; B.\\ &amp; A \times I \ar
@/_4pc/[rru]_&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;{H} \ar [r]^{f \times \id _I} &amp; X\times I \ar @{--&gt;}[ur]^{H&apos;} } \]
</p>

<p>
\[\hskip -0.5cm\xymatrix @C=25pt@R=30pt{&amp; T\times T\ar [drr]_{\sqcup } \ar [rr]^{ \tau } &amp;\drtwocell \omit {\,\,\, \mathbf {R}}&amp; T \times T \ar [d]^{\sqcup } \\ &amp;&amp;&amp; T.}
\]
</p>

<p>
The responsibility for this HTML file is <a href="https://joaofariamartins.github.io/" target="_blank" >Joa&#x0303;o Faria Martins</a>' only.<br />
Corrections, suggestions, etc. are very welcome, and can be sent to: <a href="mailto:j.fariamartins@leeds.ac.uk" target="_blank" >j.fariamartins@leeds.ac.uk</a>.<br />
<b>This is strictly a non-specialist experimental file.</b><br />

</p>

<p>
See the links below for:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">●</span> <a href="Sample_Lwarp_Xy-jax.html" target="_blank" >"stand-alone" version of this webpage</a>,
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> <a href="Sample_Lwarp_Xy-jax-split-index.html" target="_blank" >"split" version of this webpage</a> (with sans serif fonts),
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> <a href="Sample_Lwarp_Xy-jax.pdf" target="_blank" >pdf version of this page</a>,
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> Latex source for the stand alone version of this page <a href="Sample_Lwarp_Xy-jax.tex" target="_blank" >Sample_Lwarp_Xy-jax.tex</a>,
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> Latex source for the split version of this page, with sans serif fonts: <a href="Sample_Lwarp_Xy-jax-split.tex" target="_blank" >Sample_Lwarp_Xy-jax-split.tex</a>,
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> CSS files for the stand-alone version of this page: <a href="lwarp.css" target="_blank" >lwarp.css</a> (automatically provided by <em>lwarpmk</em>) and <a
href="my_file_stand-alone.css" target="_blank" >my_file_stand-alone.css</a> (minor tweak to adjust margins),
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> CSS files for the split version of this page: <a href="lwarp\_formal.css" target="_blank" >lwarp_formal.css</a> (automatically provided by <em>lwarpmk</em>) and <a
href="my_file_split.css" target="_blank" >my_file_split.css</a> (minor tweak enforcing sans serif fonts).
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> Configuration file required for browsers to find Xy-Jaxv6, <a href="lwarp-with-Xy-jax_v6.txt" target="_blank" >lwarp-with-Xy-jax_v6.txt</a>. This script is essentially the original
Lwarp MathJax emulation code, <a href="lwarp_mathjax.txt" target="_blank" >lwarp_mathjax.txt</a> (automatically created by lwarp v0.902), with two minimal additions, as shown. File tweaked 25May2026. Requires <a
href="https://raw.githubusercontent.com/sonoisa/XyJax-v3/master/build/xypic.js" target="_blank" >xypic.js</a> in directory "js".
</p>
</li>
</ul>
```

```
<!--
......  section Context and instructions:  using Lwarp to compile latex to HTML, so that xymatrix diagrams are displayed with Xy-Jax ......
-->
<h4 id="autosec-4"><span class="sectionnumber">1&#x2003;</span>Context and instructions: using Lwarp to compile latex to HTML, so that xymatrix diagrams are displayed with Xy-Jax</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-4"></a>


<a id="Context"></a>

<p>
This HTML file (that you are reading) was produced with the <a href="https://ctan.org/pkg/lwarp" target="_blank" >lwarp.sty package</a>, available with <b>TeXLive</b>. <sup>1</sup> In particular the preamble
contains:
</p>
```

```
<pre class="verbatim">
\documentclass[a4paper,12pt]{article}
\usepackage[mathjax]{lwarp}

</pre>

<p>
The version of the software was:
</p>
```

```
<pre class="verbatim">
 pdfTeX, Version 3.141592653-2.6-1.40.24 (TeX Live 2022) (preloaded format=pdflatex)
 lwarpmk: lwarpmk: v0.910  Automated make for the LaTeX Lwarp package.

</pre>

<p>
The source latex file can be found here: <a href="Sample_Lwarp_Xy-jax.tex" target="_blank" >Sample_Lwarp_Xy-jax.tex</a> or here <a href="Sample_Lwarp_Xy-jax-split.tex" target="_blank"
>Sample_Lwarp_Xy-jax-split.tex</a>, depending whether you are seeing the <a href="Sample_Lwarp_Xy-jax.html" target="_blank" >“stand-alone” version of this webpage</a>, or the <a
href="Sample_Lwarp_Xy-jax-split-index.html" target="_blank" >“split” version of this webpage</a>.
</p>

<p>

</p>

<p>
This HTML file also makes use of the following configuration script, so that <b>Xy-pic</b> diagrams, particularly <a href="https://www.ctan.org/pkg/xymatrix" target="_blank" ><i>xymatrix</i></a> commmutative
diagrams, can be displayed by using <b>Xy-Jax-v3</b>. (Or at least a good sample of them: there seem to be some compatibility issues.)<br />

</p>
```

```
<div class="center">

<p>
<a href="lwarp-with-Xy-jax_v6.txt" target="_blank" >lwarp-with-Xy-jax_v6.txt</a>.
</p>
</div>

<p>
This script file has minimal additions (two in total) in the original Lwarp MathJax emulation code, that is automatically created by lwarp v0.902), namely:
</p>
```

```
<div class="center">

<p>
<a href="lwarp_mathjax.txt" target="_blank" >lwarp_mathjax.txt</a>,
</p>
</div>

<p>
This modification was produced under items 5 and 6 of the licence <a href="https://www.ctan.org/license/lppl1.3" target="_blank" >https://www.ctan.org/license/lppl1.3</a> (The LATEX Project Public License
1.3).<br />
Credit is due to instructions in: <a href="https://github.com/sonoisa/XyJax-v3" target="_blank" >https://github.com/sonoisa/XyJax-v3</a> (the official Xy-Jaxv3 documentation) and in "40.11 lwarp_mathjax.txt" of the
<a href="http://mirrors.ibiblio.org/CTAN/macros/latex/contrib/lwarp/lwarp.pdf" target="_blank" >official Lwarp documentation</a>.
</p>

<p>
If using Xy-pic diagrams displayed with <b>XyJax-v3</b>, insert the following in the preamble of your latex file:
</p>
```

```html
<pre class="verbatim">
  \usepackage[all]{xy}
  \MathJaxFilename{lwarp-with-Xy-jax_v6.txt}

</pre>

<p>
Also create a directory “js”, and copy the file <a href="https://raw.githubusercontent.com/sonoisa/XyJax-v3/master/build/xypic.js" target="_blank" >xypic.js</a> to it.
</p>

<p>
<b>Compile .tex to .html using:</b>
</p>

<ul class="enumerate" style="list-style-type:none">

<li>
<p>
<span class="listmarker">1.</span> <em>pdflatex Sample_ Lwarp_ Xy-jax.tex</em> (possibly twice) : creates Sample_Lwarp_Xy-jax.pdf (usual pdf file, that can also be shared) and Sample_Lwarp_Xy-jax_html.tex. (And
some other aux files.)
</p>

</li>
<li>

<p>
<span class="listmarker">2.</span> <em>lwarpmk html</em> : converts Sample_Lwarp_Xy-jax_html.tex to Sample_Lwarp_Xy-jax_html.html. And then creates Sample_Lwarp_Xy-jax.html (possibly split into different
HTML files if such option is chosen).
</p>

</li>
<li>

<p>
&#x2003;&#x2003;<b>Note:</b> Use <em>lwarpmk html1</em> to force a recompile.
</p>

</li>
<li>

<p>
&#x2003;&#x2003;<b>Note:</b> In case of error, with <em>lwarpmk html</em>, use <em>lwarpmk pdftohtml</em> to create HTML file.
</p>

</li>
<li>

<p>
&#x2003;&#x2003;<b>Note:</b> When using <em>bibtex</em> files, then the following is required:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">•</span> bibtex Sample_Lwarp_Xy-jax_html.aux
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> <em>pdflatex</em> Sample_Lwarp_Xy-jax_html.tex
</p>
</li>
</ul>
</li>
<li>

<p>
The following steps are only needed once (any time tikz figures change). Only needed with tikz.
</p>

</li>
<li>

<p>
<span class="listmarker">3.</span> <em>lwarpmk limages</em> : If there are images done with tikz, tikz-cd, etc.
</p>

</li>
<li>

<p>
<span class="listmarker">4.</span> Repeat 2 (now using <em>lwarpmk html1</em>) and 3 (to finish creating the images).
</p>
</li>
</ul>
```

```
<div role="note" class="footnotes">

<a id="Sample_Lwarp_Xy-jax-autopage-10"></a>

<p>
<sup>1</sup>&nbsp;See also <a href="https://people.bath.ac.uk/feb/lwarp/lwarp-intro.html" target="_blank" >https://people.bath.ac.uk/feb/lwarp/lwarp-intro.html</a>, for a quick introduction on how to use
Lwarp, and without which this explanation that you are reading would not have been written.
</p>


</div>



<div class="amsthmbodyplain">

<ul class="list" style="list-style-type:none">



<li>

<p>

<span class="listmarker">

<span class="amsthmnameplain">Note</span>. </span> It is advisable to use a recent version of <em><b>TexLive</b>.</em>

</p>

</li>

</ul>

</div>

<p>
<b>Warnings:</b>
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">●</span> The usage of <a href="https://github.com/sonoisa/XyJax-v3" target="_blank" ><b>XyJax-v3</b></a> software with <a href="https://ctan.org/pkg/lwarp" target="_blank"
>lwarp</a> is experimental.
</p>

</li>
<li>

<p>
<span class="listmarker">●</span> Not all accessibility features of commutative diagrams displayed with XyJax-v3 work.<br />
Among the accessibility features that seem to work well are:
</p>
<ul class="enumerate" style="list-style-type:none">

<li>
<p>
<span class="listmarker">1.</span> View original latex source.
</p>

</li>
<li>

<p>
<span class="listmarker">2.</span> Zoom maths expression.
</p>
</li>
</ul>
</li>
</ul>
```

```
<!--
......  section Examples:  xymatrix diagrams displayed with Xy-Jaxv3 using Lwarp ......
-->
<h4 id="autosec-11"><span class="sectionnumber">2&#x2003;</span>Examples: xymatrix diagrams displayed with Xy-Jaxv3 using Lwarp</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-11"></a>


<a id="sec:context"></a>

<p>
This is an example of a HTML file with:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">•</span> Mathematical formulae displayed with <a href="https://www.mathjax.org/" target="_blank" ><b>MathJax</b></a>:
</p>
<span class="hidden"> \(\seteqnumber{0}{}{0}\)</span>
<!--

                                                                      x^2 + y^2 = 1,                                    (1)--><a id="test"></a><!--

-->
<p>

\begin{equation}
\label {test} x^2+y^2=1,
\end{equation}

</p>
<p>
\[\big ( x \ra { \gamma } y\big ) \in \pi _1(X,X_0), \]
</p>
<p>
\[ \dots \to \pi _i(E_b,x) \ra {\iota } \pi _i(E,x) \ra {\d } \pi _i(B,b) \ra {\delta } \pi _{i-1}(E_b,x) \to \dots \ra {\iota } \pi _1(E,x) \ra {\partial } \pi _1(B,b) \ra {\delta _x} \pi
_0(E_b) \ra {\iota } \pi _0(E)=\{0\}.  \]
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> To see how lecture notes look like with Lwarp, see subsections <a href="Sample_Lwarp_Xy-jax.html#adjunctions">6.1</a> and <a href="Sample_Lwarp_Xy-jax.html#Galois">6.2</a>.
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> This configuration is compatible with AMS-CD commutative diagrams, e.g. this diagram from the <a href="https://www.jmilne.org/not/Mamscd.pdf" target="_blank" >amscd
package manual</a>:
</p>
<p>
\[ \begin {CD} A @&gt;a&gt;b&gt; B\\ @VlVrV @AlArA\\ C @&lt;a&lt;b&lt; D \end {CD} \]
</p>
```

```
<blockquote>
</blockquote>
</li>
<li>

<p>
<span class="listmarker">•</span> Commutative diagrams drawn with <a href="https://tug.org/applications/Xy-pic/" target="_blank" ><b>Xy-pic</b></a> displayed by using <a
href="https://github.com/sonoisa/XyJax-v3" target="_blank" ><b>XyJax-v3</b></a>.<br />
For a good introduction to typeseting diagrams with Xy-pic see e.g.
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">–</span> <a href="https://www.math.purdue.edu/~adebray/lecture_notes/using_xy.pdf" target="_blank" >https://www.math.purdue.edu/~adebray/lecture_notes/using_xy.pdf</a>
(Arun Debray Using XY to typeset automata. July 5, 2016.)
</p>
</li>
<li>

<p>
<span class="listmarker">–</span> <a href="https://web.math.ku.dk/~holm/download/xyrefer.pdf" target="_blank" >https://web.math.ku.dk/~holm/download/xyrefer.pdf</a> (Kristoffer H. Rose &amp; Ross Moore:
XY-pic Reference Manual, Version 3.7 1999/02/16.)
</p>
</li>
<li>

<p>
<span class="listmarker">–</span> See also the more recent <a href="https://anorien.csc.warwick.ac.uk/mirrors/CTAN/macros/generic/diagrams/xypic/doc/xyguide.pdf" target="_blank" >user guide</a>
(Kristoffer H. Rose, Version 3.8.9, October 6, 2013) in the official page of <a href="https://ctan.org/pkg/xymatrix?lang=en" target="_blank" >xymatrix – Commutative diagrams using XY-pic </a>
</p>
</li>
</ul>
<p>
E.g.
</p>
<p>
\[\xymatrix @R=60pt@C=60pt{&amp;S_3 \ar @{^{(}-&gt;}[r]^f \ar [dr]_{g}^{\cong } &amp;S_4 \ar @{-&gt;&gt;}[d]^\pi \\ &amp;&amp;S_4/V, } \]
</p>
<p>
\[ \xymatrix { &amp;&amp; \Q \\ &amp; \Q (\w ) \ar @{&lt;-}[ur] \\ &amp; &amp; \Q (\g )\ar @{&lt;-}[uu] &amp; \Q (\b )\ar @{&lt;-}[uul] &amp; \Q (\a ),\ar @{&lt;-}[uull]\\ &amp;&amp;\Q (\a ,\b
,\g )\ar @{&lt;-}[uul]\ar @{&lt;-}[u] \ar @{&lt;-}[ur] \ar @{&lt;-}[urr] } \]
</p>
<p>
\[\xymatrix @R=1pt{\\\\\\(123).}\xymatrix @R=39pt{ &amp; \red {\bullet }\ar @{-}[dr] \ar @{-}[rr] &amp;&amp; \blue {\star }\ar @{-}[dl] \\ &amp;&amp; \green {\square } }\qquad \xymatrix
@R=1pt{\\\\\\\\ =} \xymatrix @R=39pt{ &amp; \blue {\star }\ar @{-}[dr] \ar @{-}[rr] &amp;&amp; \green {\square }\ar @{-}[dl] \\ &amp;&amp; \red {\bullet } } \xymatrix @R=1pt{\\\\\\\\ ,}\]
</p>
<p>
\[\xymatrix @R=30pt@C=60pt {&amp;A\ar [d]_{{\iota _0}^A} \ar [r]^f &amp; X\ar [d]_{{\iota _0}^X} \ar [r]^g &amp; B,\\ &amp; A \times I \ar
@/_4pc/[rru]_&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;&lt;{H} \ar [r]^{f \times \id _I} &amp; X\times I \ar @{--&gt;}[ur]^{H&apos;} } \]
</p>
<p>
\[ \xymatrix { &amp; \displaystyle \bigoplus _{y \in H_0} \mathbf {H}_1(x,y)\otimes \mathbf {H}_1&apos;(y,z)\ar [dd]_{\displaystyle \bigoplus _{y \in H_0} \eta _{(x,y)} \otimes \eta
&apos;_{(y,z)}} \ar [rr]^{\mathrm {proj}} &amp;&amp; \displaystyle \int ^{y \in H_0} \mathbf {H}_1(x,y)\otimes \mathbf {H}_1&apos;(y,z)\ar @{--&gt;}[dd]^{ \displaystyle \int ^{y \in Y} \eta
_{(x,y)} \otimes \eta &apos;_{(y,z)} }\\ \\ &amp; \displaystyle \bigoplus _{y \in H_0} \mathbf {H}_2(x,y)\otimes \mathbf {H}_2&apos;(y,z) \ar [rr]_{\mathrm {proj}} &amp;&amp; \displaystyle
\int ^{y \in H_0} \mathbf {H}_2(x,y)\otimes \mathbf {H}_2&apos;(y,z)\,\, , } \]
</p>
<p>
\[\xymatrix { &amp;\displaystyle \int ^{y \in {Y}} \Hpb _{(X,Y)}^{M_1}(-,y) \otimes \Hpb _{(Y,Z)}^{M_2}(y,-)\ar @{=&gt;}[dd]|{\tHpb {\mathbf {W}_1}{X}{Y} \bullet \tHpb {\mathbf {W}_2}{Y}{Z}}
\ar @{=&gt;}[rrr]^&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;{\eta _{(X,Y,Z)}^{M_1,M_2}}&amp;&amp;&amp; \Hpb _{(X,Z)}^{M_1\times _Y M_2} \ar @{=&gt;}[dd]|{\tHpb
{\mathbf {W}_1\#_0 \mathbf {W}_2}{X}{Z}} \\\\ &amp;\displaystyle \int ^{y \in {Y}} \Hpb _{(X,Y)}^{N_1}(-, y) \otimes \Hpb _{(Y,Z)}^{N_2}(y,-)\ar
@{=&gt;}[rrr]^&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;{\eta _{(X,Y,Z)}^{N_1,N_2}}&amp;&amp;&amp; \Hpb _{(X,Z)}^{N_1\times _Y N_2}\,.  } \]
</p>
<p>
\[\xymatrix { &amp;\Cc \rtwocell ^F_{F&apos;}{\eta } &amp;\Dc } \]
</p>
<p>
\[\hskip -0.5cm\xymatrix @C=25pt@R=30pt{&amp; T\times T\ar [drr]_{\sqcup } \ar [rr]^{ \tau } &amp;\drtwocell \omit {\,\,\, \mathbf {R}}&amp; T \times T \ar [d]^{\sqcup } \\ &amp;&amp;&amp; T.}
\]
</p>
<p>
\[\xymatrix @C=45pt{&amp;\Cc \sqcup \Cc &apos; \rtwocell \omit {\quad \,\, \langle \eta ,\eta &apos;\rangle } \ar @/^1.5pc/[r]^{T_b\circ \langle f, f&apos; \rangle } \ar @/_1.5pc/[r]_{\langle
T,T&apos;\rangle } &amp; \Grp .  }\]
</p>
<p>
\[ \xymatrix @C=60pt{\Cc \ruppertwocell ^F{\eta } \rlowertwocell _{F&apos;&apos;}{\eta &apos;} \ar [r]|{F&apos;&apos;} &amp; \Dc &amp; = &amp;\Cc \ar @/^1.5pc/[r]^{F}\ar
@/_1.5pc/[r]_{F&apos;&apos;}\rtwocell \omit {\quad \,\,\,\,\eta \#_1\eta &apos;} &amp; \Dc &amp;} \]
</p>
<p>
\[ \hskip -2cm\xymatrix @C=60pt{ &amp;\big ( \mathbf {Cob}^{(n,n+1,n+2)}\big )^3 \ar [d]_{\id \times \sqcup }\ar [dr]^{\sqcup \times \id } \drrtwocell \omit {\qquad \boldsymbol {\chi
}&apos;\times \mathcal {F}_{\mathbf {B}}} \ddtwocell \omit {&lt;-13&gt; \alpha } \ar [rr]^{\mathcal {F}_{\mathbf {B}} \times \mathcal {F}_{\mathbf {B}} \times \mathcal {F}_{\mathbf {B}}}
&amp;&amp; \big (\mathbf {Prof} \big )^3\ar [d]^{\otimes \times \id }\\ &amp;\big ( \mathbf {Cob}^{(n,n+1,n+2)})^2 \ar [d]_{\sqcup } &amp; \big ( \mathbf {Cob}^{(n,n+1,n+2)})^2 \ar
[dl]^{\sqcup } \dtwocell \omit { \boldsymbol {\chi }&apos;} \ar [r]^{\mathcal {F}_{\mathbf {B}} \times \mathcal {F}_{\mathbf {B}}} &amp; \big (\mathbf {Prof} \big )^2 \ar [dl]^{\otimes } \\
&amp; \big (\mathbf {Cob}^{(n,n+1,n+2)}\big ) \ar [r]_{\mathcal {F}_{\mathbf {B}}}&amp; \mathbf {Prof}\\ &amp; \ar @3[r]_{\boldsymbol {\omega }&apos;} &amp; \\ &amp;\big ( \mathbf
{Cob}^{(n,n+1,n+2)}\big )^3 \ar [d]_{\id \times \sqcup } \drrtwocell \omit {\quad \qquad \mathcal {F}_{\mathbf {B}}\times \boldsymbol {\chi }&apos; }\ar [rr]^{\mathcal {F}_{\mathbf {B}} \times
\mathcal {F}_{\mathbf {B}} \times \mathcal {F}_{\mathbf {B}}} &amp;&amp; \big (\mathbf {Prof} \big )^3\ddltwocell \omit {\alpha } \ar [d]^{\otimes \times \id }\ar [dl]^{\id \times \otimes }\\
&amp;\big ( \mathbf {Cob}^{(n,n+1,n+2)})^2 \ar [d]_{\sqcup } \drtwocell \omit {\,\,\boldsymbol {\chi }&apos; } \ar [r]^{\mathcal {F}_{\mathbf {B}} \times \mathcal {F}_{\mathbf {B}}} &amp; \big
( \mathbf {Prof})^2 \ar [d]^{\otimes } &amp; \big (\mathbf {Prof} \big )^2\ar [dl]^{\otimes }\\ &amp; \big (\mathbf {Cob}^{(n,n+1,n+2)}\big ) \ar [r]_{\mathcal {F}_{\mathbf {B}}}&amp; \mathbf
{Prof}.  } \]
</p>
</li>
</ul>
```

```
<!--
......  section Issues when using XyJax-v3 and Lwarp for typesetting xymatrix diagrams ......
-->
<h4 id="autosec-13"><span class="sectionnumber">3&#x2003;</span>Issues when using XyJax-v3 and Lwarp for typesetting xymatrix diagrams</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-13"></a>


<p>
Please do contact me if you know the solution to any of these issues. Remember: this is stricly a non-specialist file!
</p>

<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">●</span> The commands of the form:
</p>
```

```
<pre class="verbatim">
 \stackrel{(*)}{\implies}
</pre>
<p>
or
</p>
```

```
<pre class="verbatim">
\left ( x \xrightarrow{f} y\right)
</pre>
<p>
do not seem to work well inside
</p>
```

```
<pre class="verbatim">
 \begin{equation} \end{equation}.
</pre>
<p>
So use instead, for example,
</p>
```

```
<pre class="verbatim">
$$ \stackrel{(*)}{\implies} $$
</pre>
```

```
<pre class="verbatim">
$$ \left ( x \xrightarrow{f} y \right)$$
</pre>
<p>
Here is how it looks:
</p>
<p>
\[ \stackrel {(*)}{\implies }\]
</p>
<p>
\[\left ( x \xrightarrow {f} y \right ) .\]
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> Sometimes there are errors if there are mathematical commands in the title of sections, subsections, etc.
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> Lwarpmk gives error messages if it finds <i>xymatrix</i> inside:
</p>
```

`<pre class="verbatim">`
` \begin{equation} \end{equation}.`
`</pre>`
`<p>`
(However a html file can still be produced with `<i>`*lwarpmk pdftohtml*`</i>`.)
`</p>`
`<p>`
So avoid:
`</p>`

```
<pre class="verbatim">
 \begin{equation} \xymatrix{    } \end{equation},
</pre>
<p>
and use instead:
</p>
```

```
<pre class="verbatim">
 \[ \xymatrix{   } \],
</pre>
<p>
or
</p>
```

```
<pre class="verbatim">
 $$ \xymatrix{   } $$.
</pre>
</li>
<li>

<p>
<span class="listmarker">•</span> <b>Not all accessibility functions of MathJax work with Xy-Jaxv3, only some can be used, e.g. zoom and source latex code</b>.<br />
Accessibility features of Xy-jax <a href="https://github.com/sonoisa/XyJax-v3" target="_blank" >are not officially supported</a>.<br />

</p>
</li>
<li>

<p>
<span class="listmarker">•</span> References to section, chapters etc, have glitches <b>if inside mathematics environments</b>. For instance, the links provided do not appear to work well. E.g.
</p>
<span class="hidden"> \(\seteqnumber{0}{}{1}\)</span>
<!--
```

$$\exp(x+yi)=e^x\big(\cos(y)+i\sin(y)\big) \qquad (2)$$

```
--><a id="Euler"></a><!--
-->
<p>

\begin{equation}
\label {Euler} \exp (x+yi)=e^x\big (\cos (y)+i\sin (y) \big )
\end{equation}

</p>
<span class="hidden"> \(\seteqnumber{0}{}{2}\)</span>

<!--
```

$$\begin{aligned} e^{i\pi}+1 &= \big(\cos(\pi)+i\sin(\pi)\big)+1, \quad \text{using Equation (2). (Just for testing) equations (1), (3)} \\ &= 0. \quad \text{A reference to second section 2} \end{aligned}$$

```
-->

<p>

\begin{align*}
e^{i\pi }+1&amp;=\big (\cos (\pi )+i\sin (\pi ) \big )+1, \quad \textrm { using Equation } \eqref {Euler}.  \textrm { (Just for testing) equations } \eqref {test}, \eqref {xydiagram}\\
&amp;=0.  \quad \textrm { A reference to second section } \protect \ref {sec:context}
\end{align*}
Outside maths environments all seems to work fine:
</p>
```

```
<blockquote>
<p>
Using Equation <span class="textup">(<a href="Sample_Lwarp_Xy-jax.html#Euler">2</a>)</span>. (Just for testing) equations <span class="textup">(<a href="Sample_Lwarp_Xy-jax.html#test">1</a>)</span>,
<span class="textup">(<a href="Sample_Lwarp_Xy-jax.html#lateximage-Sample_Lwarp_Xy-jax-1">3</a>)</span><br />
A reference to second section <a href="Sample_Lwarp_Xy-jax.html#sec:context">2</a>
</p>
</blockquote>
</li>
<li>

<p>
<span class="listmarker">•</span> When compiling xymatrix, or tikz, diagrams as sgv figures, with Alt-Text (which sometimes is necessary) it sometimes happens that the figure is not cropped correctly. This seems to depend
stronly on the operating system being used.
</p>
</li>
</ul>
```

```
<!--
......  section More general notes ......
-->
<h4 id="autosec-24"><span class="sectionnumber">4&#x2003;</span>More general notes</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-24"></a>


<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">•</span> <b>pdf</b> and <b>HTML</b> automatically coexist: here is the pdf version of this HTML file <a href="Sample_Lwarp_Xy-jax.pdf" target="_blank"
>Sample_Lwarp_Xy-jax.pdf</a>
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> It is possible to split an HTML page into sub-pages: credit <a href="https://people.bath.ac.uk/feb/lwarp/lwarp-intro.html" target="_blank"
>https://people.bath.ac.uk/feb/lwarp/lwarp-intro.html</a>. Instructions can be seen in the source latex file of this HTML file <a href="Sample_Lwarp_Xy-jax.tex" target="_blank" >Sample_Lwarp_Xy-jax.tex</a>.
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> Additions are required in the .tex file in order that <b>MathJax</b> displays commands, macros and definitions correctly. E.g. write:
</p>
```

```
<pre class="verbatim">
\DeclareMathOperator{\Sym}{Sym}
\def \Mon {{\mathbf{Mon}}}
\newcommand{\Q}{\mathbb{Q}}
</pre>
<p>
and then:
</p>
```

```
<pre class="verbatim">
\begin{warpHTML}
  \CustomizeMathJax{\DeclareMathOperator{\Sym}{Sym}}
  \CustomizeMathJax{\def \Mon {{\mathbf{Mon}}}}
  \CustomizeMathJax{\newcommand{\Q}{\mathbb{Q}}}
\end{warpHTML}
</pre>
<p>
This means that you can use slightly different versions of commands for pdf and for html.
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> If using Xy-pic diagrams (if displayed as Xy-jax) put in the preamble of the tex file:
</p>
```

```
<pre class="verbatim">
 \MathJaxFilename{lwarp-with-Xy-jax_v6.txt}
</pre>
<p>
You will also need this configuration file: <a href="lwarp-with-Xy-jax_v6.txt" target="_blank" >lwarp-with-Xy-jax_v6.txt</a>. (Explanation is in Section <a href="Sample_Lwarp_Xy-jax.html#Context">1</a>. Also
requires <a href="https://raw.githubusercontent.com/sonoisa/XyJax-v3/master/build/xypic.js" target="_blank" >xypic.js</a> in directory “js”)
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> Sample use of Xy-pic compatible with lwarp (so that the resulting xymatrix code is readable by Xy-Jax): E.g.:
</p>
```

```
<pre class="verbatim">
\[
\xymatrix
{
&& \Q\\ & \Q(\w) \ar@{<-}[ur] \\
& & \Q(\g)\ar@{<-}[uu] & \Q(\b)\ar@{<-}[uul] & \Q(\a)\ar@{<-}[uull]\\
&&\Q(\a,\b,\g)\ar@{<-}[uul]\ar@{<-}[u] \ar@{<-}[ur] \ar@{<-}[urr]
}
\],
</pre>
<p>
or
</p>
```

```
<pre class="verbatim">
\displaymathnormal{$$
\xymatrix{
&S_3 \ar@{^{(}->}[r]^f \ar[dr]_{g}^{\cong} &S_4 \ar@{->>}[d]^\pi
\\ &&S_4/V,
}
$$}.
</pre>
</li>
<li>

<p>
<span class="listmarker">●</span> Figures, with alternative text, can be included like this:
</p>
```

```
<pre class="verbatim">
\includegraphics[width = 0.4\textwidth,
alt={Type your alternative text here}]{Stasheff.png}
</pre>
```

```
<div class="center">
<p>

<a href="Stasheff.png"  title="This figure shows Stasheff’s Polytope.  The top face is a pentagon, the bottom face is a pentagon ..." target="_blank" ><img
    src="Stasheff.png"
    style="
    width:130pt;
    "
    class="inlineimage"
    alt="(This figure shows Stasheff’s Polytope.  The top face is a pentagon, the bottom face is a pentagon ...)"
></a>
</p>
</div>
</li>
<li>

<p>
<span class="listmarker">•</span> Figures, including <b>tikz</b> and <b>tikz-cd figures</b>, can be compiled as figures with <b>Alt Text</b>. (The same option is also available for xymatrix diagrams, and may be
preferable to Xy-Jaxv3, in the cases when it is sensible to provide a comprehensive alternative text to the figure/diagram, instead of relying mainly on the availability of the latex code, for accessibility.)<br />
<b>Warning:</b> In some operating systems, issues seem to exist with the conversion of tikz figures to svg: e.g. figures may be incorrectly cropped for the web-page.<br />
Size of xymatrix figures and tikz-cd figures seemingly then must be adjusted manually.
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> When using <b>tikz</b> pictures use (note the Alt Text option):
</p>
```

```
<pre class="verbatim">
\begin{figure}\ThisAltText{Alt text to your diagram}
\begin{tikzpicture}

\end{tikzpicture}
\end{figure}
</pre>
<p>
or
</p>
```

```
<pre class="verbatim">
\begin{center}\ThisAltText{Alt text to your diagram}
\begin{tikzpicture}

\end{tikzpicture}
\end{center}
</pre>
<p>
Examples can be found towards the end of this file.
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> When using <b>tikz-cd</b> use:
</p>
```

```
<pre class="verbatim">
{\displaymathother
 \ThisAltText{ Alt text to your commutative diagram }
\[
    \begin{tikzcd} F(A) \arrow{r}{F(f)}& F(B) \\
    G(A)\arrow{r}[swap]{G(f)} \arrow{u}{\eta_A} & G(B)
    \arrow{u}[right]{\eta_B}
    \end{tikzcd}
\]
}
</pre>
</li>
<li>

<p>
<span class="listmarker">•</span> It is possible to compile an <b>Xy-pic diagram</b> as a <b>figure with alt text</b>. Use:
</p>
```

```
<pre class="verbatim">
{
\displaymathother
\ThisAltText{Write some alt text here.}
$$ \xymatrix{ } $$
}
</pre>
<p>
Examples can be found towards the end of this file.
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> <b>To select your own .CSS files use:</b>
</p>
```

```
<pre class="verbatim">
 \CSSFilename{your_file.css}
</pre>
</li>
<li>

<p>
<span class="listmarker">•</span> Avoid good old Tex commands like
</p>
```

```
<pre class="verbatim">
 {\bf }, {\it }.
</pre>
<p>
It seems that lwarp does not deal with them properly. Instead use:
</p>
```

```
<pre class="verbatim">
 \textbf{ }, \textit{ }.
</pre>
<p>
(In fact better to avoid <i>italics</i> altogether for accessibility reasons: use <b>bold</b>.)
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> Lwarp gives errors messages when compiling Xy-pic diagrams inside
</p>
```

```
<pre class="verbatim">
 \begin{equation} \end{equation}
</pre>
<p>
Use instead:
</p>
```

```
<pre class="verbatim">
\[\xymatrix{} \]
or
\[\begin{xy} \xymatrix{} \end{xy}\]
</pre>
<p>
Or use <em>lwarpmk pdftohtml</em> if there are compilation errors.
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> Figures created with tikz and xymatrix frequently have issues: e.g. they may be too small (so size needs to be adjusted), or incorrectly cropped. Seems to depend on operating system.
</p>
</li>
</ul>
```

```
<!--
......  section Some examples of tikz and xymatrix diagrams compiled as svg images, with alternative text ......
-->
<h4 id="autosec-41"><span class="sectionnumber">5&#x2003;</span>Some examples of tikz and xymatrix diagrams compiled as svg images, with alternative text</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-41"></a>


<p>
As written in Section <a href="Sample_Lwarp_Xy-jax.html#sec:context">2</a>, it is possible to display xymatrix diagrams by using XyJax-v3. In some cases, it may however be preferable to compile xymatrix diagrams as
figures with <b>Alt Text</b>. (E.g. it may be sensible to instead provide a comprehensive alternative text to the figure/diagram, instead of relying mainly on the availability of the latex code, for accessibility.)<br />

</p>

<p>
This option (svg image with alternative text) is also available for tikz and tikz-cd figures. Examples are below.
</p>
```

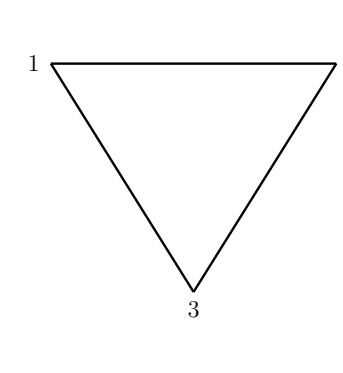
```
<!--
......  subsection An xymatrix diagram compiled as an image with alt text ......
-->
<h5 id="autosec-42"><span class="sectionnumber">5.1&#x2003;</span>An xymatrix diagram compiled as an image with alt text</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-42"></a>


<a id="xymatrix-with-alttext"></a>

<p>
An <i>xymatrix</i> diagram compiled into a picture with alt text. Can utilize equation numbers, with no errors. The size of the svg image must be adjusted. Sometimes the image is not correctly cropped: depends on operating
system.
</p>


<div class="displaymathnumbered">

<a id="autoid-1" ></a >

<p>
<span
    id="lateximage-Sample_Lwarp_Xy-jax-1"
    class="lateximagesource"
><!--
```

$$\begin{tikzcd}
T \times T \arrow[r, "\tau"] \arrow[dr, "\sqcup"', ""{name=D}] & T \times T \arrow[d, "\sqcup", ""'{name=V}] \\
 & T.
\arrow[Rightarrow, from=V, to=D, "\mathbf{R}"]
\end{tikzcd} \tag{3}$$

```
--><img
    src="Sample_Lwarp_Xy-jax-images/image-1.svg"
    alt="(This diagram shows a right triangular, diagram, with the top side horizontal, being a cathet, and containing an arrow τ, from T × T to T × T.  From the right-hand-end of the cathet,
there is vertical downwards arrow (also a cathet), ⊔: T × T → T.  From the left-hand-side of the initial cathet, there is the hypothese downwards arrow marked with ⊔: T × T → T.  There is a
2-cell from the composition of the two cathets to the hypothese, marked with boldface R)"
    role="math"
    style=""
    class="lateximage"
></span>
</p>
</div>
```

```
<!--
......  subsection A tikz image compiled as an image with alt text:  ......
-->
<h5 id="autosec-45"><span class="sectionnumber">5.2&#x2003;</span>A tikz image compiled as an image with alt text:</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-45"></a>


<p>
Size can be adjusted. Sometimes (depending on operating system, it seems) the image may be incorrectly cropped. See figure <a href="Sample_Lwarp_Xy-jax.html#fig:triangle">1</a>.
</p>
```

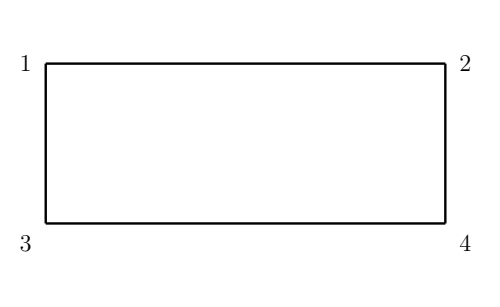
```
<div class="center">

<figure id="autoid-2" class="figure ">

<p>
<span
    id="lateximage-Sample_Lwarp_Xy-jax-2"
    class="lateximagesource"
><!--
```



```
--><img
    src="Sample_Lwarp_Xy-jax-images/image-2.svg"
    alt="(An equilateral triangle, with one of the edges horizontal.  Vertices are labelled 1, 2 and 3, clockwise, starting from the top left vertex.)"
    style=""
    class="lateximage"
></span>
</p>


<div class="figurecaption">
Figure&nbsp;1: An equilateral triangle with its vertices labeled 1, 2 and 3

</div>

<a id="fig:triangle"></a>

</figure>

</div>
```

```
<!--
......  subsection A tikz image compiled to an image with alt text, outside a float environment ......
-->
<h5 id="autosec-49"><span class="sectionnumber">5.3&#x2003;</span>A tikz image compiled to an image with alt text, outside a float environment</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-49"></a>
```

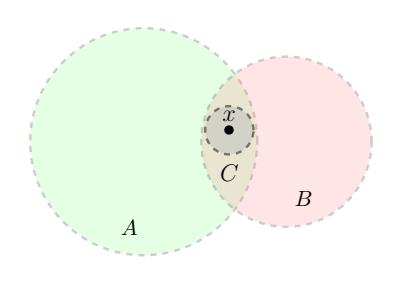
```
<div class="center">

<p>
<span
    id="lateximage-Sample_Lwarp_Xy-jax-3"
    class="lateximagesource"
><!--
```



```
--><img
    src="Sample_Lwarp_Xy-jax-images/image-3.svg"
    alt="(This figure shows a rectangle, whose largest side is horizontal.  Is vertices are labelled 1, 2 (top edge) and 3 and 4.  (bottom edge), from left to right.)"
    style=""
    class="lateximage"
></span>
</p>
</div>
```

```
<!--
......  subsection A tikz image compiled to an image with alt text, outside a float environment, and with a caption ......
-->
<h5 id="autosec-53"><span class="sectionnumber">5.4&#x2003;</span>A tikz image compiled to an image with alt text, outside a float environment, and with a caption</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-53"></a>
```

```
<div class="center">

<p>
<span
    id="lateximage-Sample_Lwarp_Xy-jax-4"
    class="lateximagesource"
><!--
```


x
C
B
A

```
--><img
    src="Sample_Lwarp_Xy-jax-images/image-4.svg"
    alt="(The figure shows two disks, A and B, of the same radius, centered at points, on a horizontal line.  The radii are smaller than the distance between the center points, but greater
than half of the distance.  The intersection of the two disks is called C.  There is a point x ∈ A ∩ B, and another disk, D, centered in x and contaned in A ∩ B.  )"
    style=""
    class="lateximage"
></span>
</p>

<div class="figurecaption">
<a id="autoid-3" ></a >
Figure&nbsp;2: Given any \(A,B \in \mathcal {F}\), if \(x \in A \cap B\) there exists \(C \in \mathcal {F}\) such that \(x\in C\subseteq A \cap B\). <a id="fig:basis"></a>
</div>
</div>
```

```html
<!--
......  subsection A tikz-cd diagram compiled into a figure with alt text:    ......
-->
<h5 id="autosec-57"><span class="sectionnumber">5.5&#x2003;</span>A tikz-cd diagram compiled into a figure with alt text: </h5>
<a id="Sample_Lwarp_Xy-jax-autopage-57"></a>


<p>
Issues with size of diagrams created with tikz-cd. Sometimes (depending on operating system) the image may be incorrectly cropped.
</p>

<p>
The following diagram commutes.<br />

</p>


<div class="displaymath">

<a id="autoid-4" ></a >

<p>
<span
    id="lateximage-Sample_Lwarp_Xy-jax-5"
    class="lateximagesource"
><!--
```

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] & F(B) \\
G(A) \arrow[u, "\eta_A"] \arrow[r, "G(f)"'] & G(B) \arrow[u, "\eta_B"']
\end{tikzcd}$$

```
--><img
    src="Sample_Lwarp_Xy-jax-images/image-5.svg"
    alt="(A diagram showing $F(f) \circ \eta_A = \eta_B \circ G(f)$)"
    role="math"
    style=""
    class="lateximage"
></span>
</p>
</div>
```

```
<!--
......  section Sample mathematics ......
-->
<h4 id="autosec-60"><span class="sectionnumber">6&#x2003;</span>Sample mathematics</h4>
<a id="Sample_Lwarp_Xy-jax-autopage-60"></a>
```

```html
<!--
......  subsection Adjoint functors and coadjoint functors via universal arrows ......
-->
<h5 id="autosec-61"><span class="sectionnumber">6.1&#x2003;</span>Adjoint functors and coadjoint functors via universal arrows</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-61"></a>


<a id="adjunctions"></a>

<p>
Let \(\Cc \) and \(\Dc \) be categories. Let \(G\colon \Dc \to \Cc \) be a functor.
</p>
```

```
<div class="amsthmbodyplain">

<ul class="list" style="list-style-type:none">


<li>
<p>
<span class="listmarker"><a id="Sample_Lwarp_Xy-jax-autopage-62"></a>
<span class="amsthmnameplain">Definition</span><span class="amsthmnumberplain"> <span class="textup">1</span></span>. </span> <a id="def:Uarrow"></a>Let \(A\) be an object of \(\Cc \). A <b>universal
arrow</b> from \(A\) to \(G\colon \Dc \to \Cc \) is a pair:
</p>

<p>
\[ \left (F_A, A \ra {\eta _A} G(F_A)\right ), \]
</p>

<p>
where \(F_A\) is an object in \(\Dc \) and \(\eta _A\colon A \to G(F_A)\) is an arrow in \(\Cc \), such that the following universal property is satisfied:
</p>

<p>

</p>
```

```
<div class="minipage" style="vertical-align:bottom ; justify-content:flex-start ; width:74pt ; ">

<a id="Sample_Lwarp_Xy-jax-autopage-63"></a>

</div>
```

```
<div class="minipage" style="vertical-align:bottom ; justify-content:flex-start ; width:421pt ; ">

<a id="Sample_Lwarp_Xy-jax-autopage-64"></a>

<p>
For any object of \(B\) of \(\Dc \) and any arrow \(f\colon A \to G(B)\), in \(\Cc \), there exists a <b>unique</b> arrow \(\hat {f}\colon F_A \to B\), in \(\Dc \), that makes the following diagram, in \(\Cc \),
commute:
</p>

<p>
\[\xymatrix {&amp; A \ar [drr]_f\ar [rr]^{\eta _A} &amp;&amp; G(F_A) \ar @{--&gt;}[d]^{G(\hat {f})}\\ &amp;&amp;&amp; G(B)} \]
</p>

</div>

</li>

</ul>

</div>
```

```
<div class="amsthmbodyplain">

<ul class="list" style="list-style-type:none">


<li>
<p>
<span class="listmarker"><a id="Sample_Lwarp_Xy-jax-autopage-65"></a>
<span class="amsthmnameplain">Exercise</span><span class="amsthmnumberplain"> <span class="textup">2</span></span>. </span> In the conditions of the previous definition, prove that if \(\left (F_A, A \ra
{\eta _A} G(F_A)\right )\) is a universal arrow from \(A\) to \(G\), then we have a bijection:
</p>

<p>
\[\phi _{A,B}\colon \hom _\Cc \big (A,G(B)\big ) \longrightarrow \hom _\Dc (F_A,B), \]
</p>

<p>
such that
</p>

<p>
\[ \big (f\colon A \to G(B)\big ) \stackrel {\phi _{(A,B)}}{\longmapsto } (\hat {f}\colon F_A \to B\big ).\]
</p>

<p>
Moreover, prove that the bijection \(\phi _{A,B}\) is natural in \(B\). This means that given any arrow \(g\colon B \to C\), in \(\Dc \), the following diagram (in the category of sets) commutes:
</p>

<p>
\[ \xymatrix { &amp; \hom _\Cc (A,G(B)) \ar [d]_{m\mapsto G(g)\circ m } \ar [rrrr]^{\phi _{A,B}} &amp;&amp;&amp;&amp; \hom _\Dc (F_A,B) \ar [d]^{n\mapsto g\circ n } \\ &amp; \hom _\Cc (A,G(C))
\ar [rrrr]_{\phi _{A,C}} &amp;&amp;&amp;&amp; \hom _\Dc (F_A,C)\\ } \]
</p>

<p>

</p>

</li>

</ul>

</div>
```

```
<!--
......  subsection The Galois correspondence for , over the rational field ......
-->
<h5 id="autosec-66"><span class="sectionnumber">6.2&#x2003;</span>The Galois correspondence for \(f(t)=t^3-2\), over the rational field</h5>
<a id="Sample_Lwarp_Xy-jax-autopage-66"></a>


<a id="Galois"></a>

<p>
Let \(f(t)=t^3-2 \in \Q [t]\). Let \(\w =e^{\frac {2\pi i}{3}}\). Let \(\a =\sqrt [3]{2}\), \(\beta =\a \w \) and \(\gamma =\a \w ^2\). Hence the set of roots of \(f\) is \(\{\a ,\b ,\g \}\). The splitting field of
\(f\), over \(\Q \) is:
</p>

<p>
\[\Q (\a ,\b ,\g )=\Q (\a ,\w ).\]
</p>

<p>
The monomorphism of groups,
</p>
<span class="hidden"> \(\seteqnumber{0}{}{3}\)</span>


<!--
```

$$\begin{aligned}\theta\colon \Gamma(f,\mathbb{Q}) &\longrightarrow \mathrm{Sym}(\{\alpha,\beta,\gamma\})\\ \tau &\longmapsto \left(\begin{matrix}\{\alpha,\beta,\gamma\} & \longrightarrow & \{\alpha,\beta,\gamma\}\\ a & \longmapsto & \tau(a)\end{matrix}\right),\end{aligned}$$

```
-->


<p>

\begin{align*}
\theta \colon \Gamma (f,\Q ) &amp;\longrightarrow \Sym (\{\a ,\b ,\g \})\\ \tau &amp; \longmapsto \left ( \begin{CD} \{\a ,\b ,\g \} &amp;\longrightarrow &amp;\{\a ,\b ,\g \}\\ a
&amp;\longmapsto &amp;\tau (a) \end {CD}\right ),
\end{align*}
is, in this case, an isomorphism. The diagram of subgroups of \(\Gamma (f,\Q ) \cong \Sym (\{\a ,\b ,\g \})\) is below (note that inclusions go in the direction of arrows):
</p>

<p>
\[\xymatrix { &amp;&amp; \Sym (\{\a ,\b ,\gamma \} )\\ &amp; \{\id , (\a \b \g ), (\a \g \b )\} \ar [ur] \\ &amp; &amp; \{\id ,(\a \b )\}\ar [uu] &amp;\{\id ,(\a \g )\}\ar [uul] &amp;\{\id ,
(\b \g )\}\ar [uull]\\ &amp;&amp; \{\id _R\}\ar [uul]\ar [u] \ar [ur] \ar [urr] } \]
</p>

<p>
The corresponding diagram of intermediate fields \(\Q \subseteq L \subseteq \Q (\a ,\b ,\g )\) is:
</p>

<p>
\[\xymatrix { &amp;&amp; \Q \\ &amp; \Q (\w ) \ar @{&lt;-}[ur] \\ &amp; &amp; \Q (\g )\ar @{&lt;-}[uu] &amp; \Q (\b )\ar @{&lt;-}[uul] &amp; \Q (\a )\ar @{&lt;-}[uull]\\ &amp;&amp;\Q (\a ,\b
,\g )\ar @{&lt;-}[uul]\ar @{&lt;-}[u] \ar @{&lt;-}[ur] \ar @{&lt;-}[urr] } \]
</p>

<p>
Also note that \(\{\id _R\}\), \(\Sym (R)\) and \({\{\id , (\a \b \g ), (\a \g \b )\}}\cong A_3\) are the only normal subgroups of \(\Sym (R)\). So, respectively,
</p>

<span class="hidden"> \(\seteqnumber{0}{}{3}\)</span>

<!--
```

$$\mathbb{Q}(\alpha,\beta,\gamma):\mathbb{Q},\qquad \mathbb{Q}:\mathbb{Q}\quad\text{and}\quad \mathbb{Q}(\omega):\mathbb{Q} \qquad (4)$$

```
--><a id="normalext"></a><!--

-->

<p>

\begin{equation}
\label {normalext} \Q (\a ,\b ,\g ):\Q ,\qquad \Q :\Q \quad \textrm { and } \quad \Q (\w ):\Q
\end{equation}

</p>

<p>
are the only normal extensions of \(\Q \) contained in \(\Q (\a ,\b ,\g )\). Note that the fact that the extensions in <span class="textup">(<a href="Sample_Lwarp_Xy-jax.html#normalext">4</a>)</span> are are
normal also follows from the fact that they are the splitting fields of \(p(t)=t^3-2\), \(q(t)=t\) and \(r(t)=t^2+t+1\) (which has \(\w \) and \(\w ^2\) as roots), over \(\Q \).
</p>
```

```
<div class="amsthmbodyplain">

<ul class="list" style="list-style-type:none">


<li>
<p>
<span class="listmarker"><a id="Sample_Lwarp_Xy-jax-autopage-67"></a>
<span class="amsthmnameplain">Remark</span><span class="amsthmnumberplain"> <span class="textup">3</span></span>. </span> Also note that we have a series of subfields of \(\C  \) (we use \(\leq  \) to denote
subfield):
</p>

<p>
\[\Q  \leq  \Q  (\w  )  \leq  \Q  (\a  ,\b  \,\g  )=\Q  (\a  ,\w  ).\]
</p>

<p>
Note that:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">•</span> \(\Q  (\w  ):   \Q  \) is a normal extension (since it is the splitting field of \(t^2+t+1\), over \(\Q  \)).
</p>

</li>
<li>

<p>
<span class="listmarker">•</span> \(\Q  (\w  ,\a  ):\Q  (\w  )\) is also a normal extension. This is because \(\Q  (\w  ,\a  )\) is the splitting field of \(t^3-2\) over \(\Q  (\w  )\).
</p>
</li>
</ul>

<p>
And then it follows that:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">•</span> \(\Gamma  \big  (\Q  (\w  ,\a  ),\Q  (\w  )\big  )\) is normal in \(\Gamma  (\Q  (\w  ,\a  ):   \Q  )\),
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> we have a series of subgroups of \(\Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  \big  )\):
</p>
<p>
\[\{e\}=\Gamma  \big  (  \Q  (\w  ,a):   \Q  (\w  ,\a  )  \big  )\trianglelefteq  \Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  (\w  )  \big  )  \trianglelefteq  \Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  \big  ),\]
</p>
</li>
<li>

<p>
<span class="listmarker">•</span> The quotient groups can be explicitly determined:
</p>
<ul class="itemize" style="list-style-type:none">

<li>
<p>
<span class="listmarker">–</span> \(\Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  \big  )  /  \Gamma  \big  (  \Q  (\w  ,a):   \Q  (\w  )  \big  )  \cong  \Gamma  \big  (\Q  (\w  ):   \Q  )\cong  \Z  _2\).
</p>
<p>
Where the last equation follows since \(\Q  (\w  ):   \Q  \) is a normal extension of degree 2, since it is the splitting field of the irreducible polynomial \(t^2+t+1\), over \(\Q  \). This \(\Z  _2\) is an abelian group.
</p>
</li>
<li>

<p>
<span class="listmarker">–</span> \(\Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  (\w  )  \big  )  /  \Gamma  \big  (  \Q  (\w  ,\a  ):   \Q  (\w  ,\a  )  \big  )  \cong  \Gamma  \big  (\Q  (\w  ,\a  ):   \Q  (\w  ))\cong  \Z  _3.   \)
</p>
<p>
Where the last equation follows since \(\Q  (\w  ,\a  ):   \Q  (\w  )\) is a normal extension of degree 3. This is because it is the splitting field of the polynomial \(t^3-2\), which is irreducible over \(\Q  (\w  )\). This is an abelian
group.
</p>
</li>
</ul>
<p>
What was just shown is a general patern that exists any time we compute the splitting field of a polynomial that is soluble by radicals.
</p>
</li>
</ul>

</li>

</ul>

</div>

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